

Chapter 2

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June 06, 2024

2.6

Claim 1. *For any summable function f , the Fourier transform*

$$\hat{f}(x) = \int f(x)e^{-2\pi inx} dx$$

exists as an ordinary Lebesgue integral and enjoys the following properties

1. $\|\hat{f}\|_{L^\infty} \leq \|f\|_{L^1}$
2. $f \in \mathbf{C}(\mathbb{R}^1)$
3. $\lim_{|n| \rightarrow \infty} \hat{f}(n) = 0$

Proof. 1.

$$\begin{aligned} \|\hat{f}\|_{L^\infty} &= \max_{n \in \mathbb{R}} |\hat{f}(n)| = \max_{n \in \mathbb{R}} \left| \int_{-\infty}^{\infty} f(x)e^{-2\pi inx} dx \right| \\ &\leq \max_{n \in \mathbb{R}} \int_{-\infty}^{\infty} |f(x)e^{-2\pi inx}| dx \\ &= \int_{-\infty}^{\infty} |f(x)| dx \\ &= \|f\|_{L^1} \end{aligned}$$

2.

$$\begin{aligned} |\hat{f}(m) - \hat{f}(n)| &= \left| \int_{-\infty}^{\infty} f(x)(e^{-2\pi imx} - e^{-2\pi inx}) dx \right| \\ &\leq \int |f(x)| |e^{-2\pi imx} - e^{-2\pi inx}| dx \rightarrow 0 \end{aligned}$$

as $|m - n| \rightarrow 0$ because $\lim_{m \rightarrow n} |f(x)| |e^{-2\pi imx} - e^{-2\pi inx}| = 0$ and $|e^{-2\pi imx} - e^{-2\pi inx}| < \sqrt{2}$ and since $f(x)$ is summable $|f(x)| < M$ thus $|f(x)| |e^{-2\pi imx} - e^{-2\pi inx}| < M\sqrt{2}$ for any choice of m and n . Thus by dominated convergence $|\hat{f}(m) - \hat{f}(n)| \rightarrow 0$. Thus $\hat{f} \in \mathbf{C}(\mathbb{R})$

3. Since

$$\hat{f}(\xi) = - \int f(x) e^{-2\pi i \xi(x-y)} dx = - \int f(x+y) e^{-2\pi i \xi x} dx$$

for $y = \frac{1}{2\xi}$.

$$\begin{aligned} \hat{f}(\xi) &= \frac{1}{2} \left[\int f(x) e^{-2\pi i \xi x} dx - \int f(x+y) e^{-2\pi i \xi x} dx \right] \\ &= \frac{1}{2} \int e^{-2\pi i \xi x} (f(x) - f(x+y)) dx \end{aligned}$$

Thus,

$$\left| \hat{f}(\xi) \right| = \left| \frac{1}{2} \int e^{-2\pi i \xi x} (f(x) - f(x+y)) dx \right| \leq \frac{1}{2} \int |e^{-2\pi i \xi x} (f(x) - f(x+y))| dx = \frac{1}{2} \|f - f_y\|_{L^1}$$

As $|\xi| \rightarrow \infty \implies y \rightarrow 0 \implies \frac{1}{2} \|f - f_y\|_{L^1} \rightarrow 0$ by continuity of translation. Thus $\left| \hat{f}(\xi) \right| \rightarrow 0$ as $|\xi| \rightarrow \infty$

□